Poursaeed *et al. BMC Cancer* (2024) 24:1581
https://doi.org/10.1186/s12885-024-13320-4

BMC Cancer

SYSTEMATIC REVIEW

Open Access

Survival prediction of glioblastoma patients using machine learning and deep learning: a systematic review

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Abstract

Glioblastoma Multiforme (GBM), classified as a grade IV glioma by the World Health Organization (WHO), is a prevalent and notably aggressive form of brain tumor derived from glial cells. It stands as one of the most severe forms of primary brain cancer in humans. The median survival time of GBM patients is only 12–15 months, making it the most lethal type of brain tumor. Every year, about 200,000 people worldwide succumb to this disease. GBM is also highly heterogeneous, meaning that its characteristics and behavior vary widely among different patients. This leads to different outcomes and survival times for each individual. Predicting the survival of GBM patients accurately can have multiple benefits. It can enable optimal and personalized treatment planning based on the patient's condition and prognosis. It can also support the patients and their families to cope with the possible outcomes and make informed decisions about their care and quality of life. Furthermore, it can assist the researchers and scientists to discover the most relevant biomarkers, features, and mechanisms of the disease and to design more effective and personalized therapies. Artificial intelligence methods, such as machine learning and deep learning, have been widely applied to survival prediction in various fields, such as breast cancer, lung cancer, gastric cancer, cervical cancer, liver cancer, prostate cancer, and covid 19. This systematic review summarizes the current state-of-the-art methods for predicting glioblastoma survival using different types of input data, such as clinical features, molecular markers, imaging features, radiomics features, omics data or a combination of them. Following PRISMA guidelines, we searched databases from 2015 to 2024, reviewing 107 articles meeting our criteria. We analyzed the data sources, methods, performance metrics and outcomes of the studies. We found that random forest was the most popular method, and a combination of radiomics and clinical data was the most common input data.

Keywords Glioblastoma multiforme (GBM), Survival prediction, Deep learning, Machine learning, Radiomics, Systematic review

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Introduction

A brain tumor is an abnormal tissue mass in which cells grow and multiply uncontrollably. Brain tumors are classified into two categories based on their malignancy and growth rate. The first category is primary, and the second is secondary. Primary brain tumors originate from the brain's tissues or the brain's immediate surroundings. In contrast, secondary tumors have spread (metastasized) from where they first started to another body part [1].

Gliomas are a kind of primary brain tumor that occur when the glial cells multiply abnormally. Gliomas grow in the brain and can also affect the spinal cord. Gliomas are divided into four grades (I, II, III, and IV) based on the World Health Organization (WHO) classification, and the grade determines the treatment and prognosis. Grade I and II gliomas are called low-grade gliomas (LGG), and Grade III and IV gliomas are called high-grade gliomas (HGG). The LGG group (also known as the benign tumor) has a longer survival rate than the HGG group.

The most aggressive type of primary brain tumor is Grade IV glioma, also known as glioblastoma multiforme (GBM) [2]. Traditional treatment of GBM is surgical resection followed by radiation therapy or chemotherapy. However, GBM is still associated with poor prognosis, and patients' median overall survival remains limited to 12–15 months [3, 4]. The inherent heterogeneity of GBM usually causes variations in prognosis, making the overall survival time largely varied across individuals even though the patients have the same tumor grade and receive similar treatments [5, 6]. Accurate prediction of prognosis and survival for each patient is crucial for patients and their families and physicians to support personalized treatment, identify who benefits from aggressive or moderate treatment and avoid ineffective treatment. For example, suppose a patient might have a short survival period. In that case, they probably will not be able to benefit from the surgical treatment and is suitable for palliative irradiation or supportive care. The patient can have surgery if he has a good prognosis and is in general condition [7–9].

Survival prediction is usually done from clinical features using statistical methods. However, with the advancement of artificial intelligence methods that include machine learning and deep learning, researchers have been eager to use these methods for survival prediction, which can combine pathology, histology, molecular, imaging, and clinical features. This systematic review aims to compare studies that used machine learning and deep learning techniques for predicting the survival of GBM patients.

Method and materials

This section deals with rules, methods and search criteria used to retrieve the research works related to survival analysis of patients with Glioblastoma cancer.

Research questions

The study is carried out to find the answers for the research questions listed below:

1) What are the factors affecting the prediction of survival in patients with Glioblastoma cancer?
2) Which machine learning and deep learning models are used in the prediction of survival in patients with Glioblastoma cancer?
3) How effective are learning methods based on accuracy?
4) Which data sets are used in the prediction of survival in patients with Glioblastoma cancer?
5) What types of data are used?

Search strategy

An electronic search following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines was used to review the studies. In the first step, Pubmed, Nature, ScienceDirect, Springer, Wiley, MDPI, and Google Scholar databases with keywords: "Glioblastoma" and "Deep learning" and "Machine learning" and "Survival prediction" in the period January 1, 2015 to October 3, 2024 to retrieve articles those carried out the work related to predicting the survival of patients with glioblastoma cancer using machine learning or deep learning are queried. Table 1 shows the number of articles collected from each publisher in the first step of the search.

The next step was to refine the selected articles. Figure 1 shows the refinement and selection process of related articles to the topic under investigation, summarized in a flowchart. From a total of 3155 articles, the articles were excluded for various reasons after screening

Table 1 The number of studies found in each database

Databases	Result
PubMed	26
Nature	29
ScienceDirect	73
MDPI	87
Wiley	297
Springer	576
Google Scholar	2067

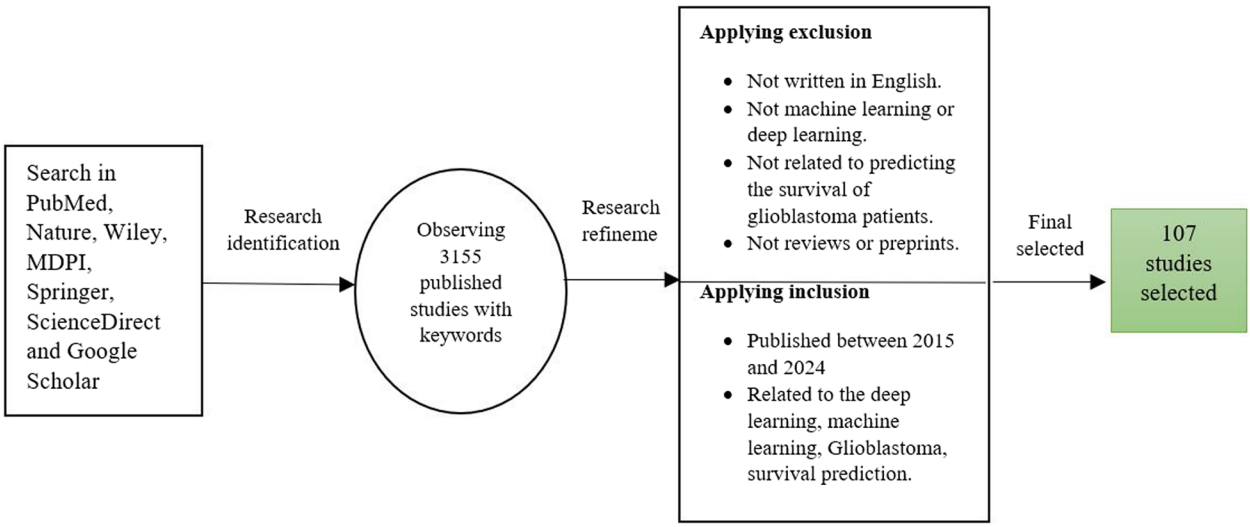


Fig. 1 The process of selecting studies

the title and abstract. The deletion index for removing them is as follows:

- Articles are not written in English.
- Preprint articles.
- Articles do not use machine learning or deep learning models.
- Articles are not related to predicting the survival of glioblastoma patients.
- Review articles.

In the refinement stage, we removed the articles that:

- Was just an abstract, poster, book chapter, thesis, report or conference proceedings.
- Duplicate articles.
- The full text of the article could not be obtained.

and articles that considered other types of cancer in addition to GBM were included in this review, only those that reported the performance value of the model separately for GBM.

The selection index for the finalized studies was based on:

- Published between 2015 and 2024.
- Related to machine learning, deep learning, glioblastoma, and survival prediction.

Finally, 107 studies were included in our systematic review.

The number of articles retrieved from different publishers after screening is shown in Fig. 2.

The selected studies are listed by publisher, title, author's name, and year of publication in Table 2.

Figure 3 shows the distribution of the papers obtained by publication year from 2015 until 2024. In 2021, the published papers had the highest rank.

Table 3 shows the journals in which the selected articles were published, the related databases and the number of articles published in each journal.

Figure 4 shows the articles retrieved from different publishers. According to it, Springer publishes 17% of articles, MDPI publishes 12%, and IEEE publishes 10% and Elsevier publishes 11%. These four publishers account for the largest share.

Findings and results

Medical background

GBM is a very aggressive form of brain cancer because it grows quickly, does not respond well to treatment, and has a low survival rate. The current treatment involves removing the tumor surgically if possible, then giving radiotherapy along with temozolomide (TMZ) regimen; however, the tumor often comes back and becomes resistant/adapted to the treatment. The bad prognosis is related to many factors such as tumor diversity, weak immune response, invasion of the tumor into "normal" tissue, production of cancer stem cells, and the fast adaptation of the tumor to harsh treatment [10]. Survival analysis is a popular topic in clinical research. Survival analysis is a branch of statistics that aims to estimate the time until an event of interest happens, giving the probability of the event happening at each time point.

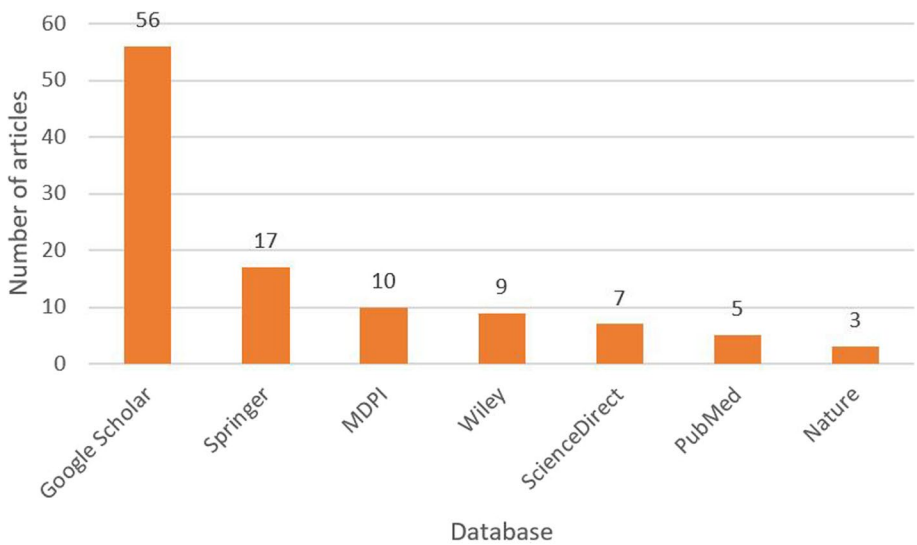


Fig. 2 Number of articles retrieved from different databases after screening

Therefore, several concepts related to the survival chance of a cancer patient are often used when a prognosis is given. Some of them are:

(a) *The risk of recurrence (e.g., local recurrence or distant metastasis), which is the chance that a treated cancer will come back,*
(b) *Cancer-specific survival (CSS), defined as the time from the diagnosis until death because of a specific type of cancer,*
(c) *Progression-free survival (PFS), defined as the time after treatment when the disease, which could not be removed, does not get worse,*
(d) *Diseasefree survival (DFS), which is the time after the disease is removed when no disease can be found and*
(e) *Overall survival (OS), defined as the time from diagnosis to death or last follow-up, with no limitation on the cause of death* [11].

In current clinical practice, clinicians usually estimate the survival, based on their experience, by looking at the images and taking into account other clinical factors such as age, gender. However, researches show that these judgments and estimates are too biased and optimistic [12]. Recent studies aim to provide better survival estimates based on multiple types of data such as clinical, molecular, and imaging data. A combination of multiple types of data may give a more detailed description of the underlying characteristics that affect survival and their relationships than single methods. For this purpose, clinical data (such as age, sex, histological grade, stage, etc.) and molecular data (such as mutation status of relevant genes, expression of proteins determined by IHC, etc.), omics data (such as genomics, epigenomics, transcription, proteomics, etc.) and imaging data (such as histological images, magnetic resonance images, computed tomography scans) are used [11].

Computational approaches

Survival prediction is a challenging task that requires the integration of multiple data sources to capture the complexity and heterogeneity of the disease. Traditional statistical approaches, such as the Cox proportional hazards model [13] and Kaplan–Meier [14] estimator, often prove inadequate for accurately predicting survival outcomes. These methods struggle when applied to high-dimensional, non-linear, and noisy datasets, particularly those derived from imaging and multi-omics data. As a result, there is a growing need for advanced computational methods that leverage artificial intelligence (AI) techniques, including machine learning (ML) and deep learning (DL), to improve the accuracy and reliability of survival prediction models. Artificial Intelligence involves using computer systems to simulate human intelligence by analyzing data and applying rules to make predictions. Machine Learning, as a branch of AI, uses data to improve performance across various tasks. Deep Learning, a subset of ML, is based on artificial neural networks with multiple layers, allowing it to process complex patterns in data. These techniques are applied to patient data to develop models that predict outcomes like survival. Data collected from large-scale institutional and registry sources—ranging from clinical records to imaging datasets—are often used to train these algorithms, enhancing their accuracy in predicting survival outcomes [15].

Table 2 Some information about the extracted studies

Database	Publisher	Name	Author	Year
ScienceDirect	Elsevier	Overall survival prediction in glioblastoma multiforme patients from volumetric, shape and texture features using machine learning	Sanghani, P., Ang, B. T., King, N. K. K., & Ren, H	2018
		Machine learning-based radiomic, clinical and semantic feature analysis for predicting overall survival and MGMT promoter methylation status in patients with glioblastoma	Lu, Y., Patel, M., Natarajan, K., Ughratdar, I., Sanghera, P., Jena, R., Watts, C., & Sawlani, V	2020
		Identification of glioblastoma molecular subtype and prognosis based on deep MRI features	Su, R., Li, X., Jin, Q., Liu, X., & Wei, L	2021
		Deep Learning Approaches for Glioblastoma Prognosis in Resource-Limited Settings: A Study Using Basic Patient Demographic, Clinical, and Surgical Inputs	Ghanem, M., Ghaith, A. K., Zamanian, C., Bon-Nieves, A., Bhandarkar, A. R., Bydon, M., & Quinones-Hinojosa, A	2023
		Prognosis prediction for glioblastoma multiforme patients using machine learning approaches: Development of the clinically applicable model	Kim, Y., Kim, K. H., Park, J., Yoon, H. I., & Sung, W	2023
		The cross-modality survival prediction method of glioblastoma based on dual-graph neural networks	Sun, J., & Peng, Y	2024
		SurvNet: A low-complexity convolutional neural network for survival time classification of patients with glioblastoma	Lyu, Q., Parreno-Centeno, M., Papa, J. P., Öztürk-Isik, E., Booth, T. C., & Costen, F	2024
Google Scholar	American Society of Neuroradiology	Deep Transfer Learning and Radiomics Feature Prediction of Survival of Patients with High-Grade Gliomas	Han, W., Qin, L., Bay, C., Chen, X., Yu, K., Miskin, N., Li, A., Xu, X., & Young, G. S	2019
	BioMed Central	Cancer survival prediction by learning comprehensive deep feature representation for multiple types of genetic data	Hao, Y., Jing, XY. & Sun, Q	2023
	British Institute of Radiology	A quantitative study of shape descriptors from glioblastoma multiforme phenotypes for predicting survival outcome	Chaddad, A., Desrosiers, C., Hassan, L., & Tanougast, C	2016
	Cukurova University Faculty of Medicine	The success of machine learning algorithms developed with radiomic features obtained from preoperative contrast-enhanced MRI in the prediction of short-term survival in patients with Glioblastoma	Dilek, O., Demirel, E., Bilgin, E., Duman, B. B., & Gülek, B	2021
	Elsevier	An Automatic Deep Learning – Based Workflow for Glioblastoma Survival Prediction Using Preoperative Multimodal MR Images: A Feasibility Study	Fu, J., Singhrao, K., Zhong, X., Gao, Y., Qi, S. L., Yang, Y., Ruan, D., & Lewis, J. S	2021
		A neuro evolutionary algorithm for patient calibrated prediction of survival in Glioblastoma patients	Ebrahimi Zade, A., Shahabi Haghighi, S., & Soltani, M	2021
		Glioblastoma Multiforme Prognosis: MRI Missing Modality Generation, Segmentation and Radiogenomic Survival Prediction	Islam, M., Wijethilake, N., & Ren, H	2021
		Dynamic architecture based deep learning approach for glioblastoma brain tumor survival prediction	Wankhede, D. S., & Rangasamy, S	2022
		Identifying overall survival in 98 glioblastomas using VASARI features at 3T	Bilmez, B. S., Firat, Z., Topcuoglu, O. M., Yaltirik, K., Türe, U., & Ozturk-Isik, E	2023

Table 2 (continued)

Database	Publisher	Name	Author	Year
	Frontiers	A Multi-parametric MRI-Based Radiomics Signature and a Practical ML Model for Stratifying Glioblastoma Patients Based on Survival Toward Precision Oncology	Osman, A. F. I	2019
		Feasibility of Multimodal MRI-Based Deep Learning Prediction of High Amino Acid Uptake Regions and Survival in Patients With Glioblastoma	Jeong, J., Lee, M. H., John, F., Robinette, N., Amit-Yousif, A. J., Barger, G., Mittal, S., & Juhász, C	2019
		Feature-Guided Deep Radiomics for Glioblastoma Patient Survival Prediction	Shboul ZA, Alam M, Vidyaratne L, Pei L, Elbakary MI and Iftekharuddin KM	2019
		Robustness of Radiomics for Survival Prediction of Brain Tumor Patients Depending on Resection Status	Weninger L, Haarburger C and Merhof D	2019
		Overall Survival Prediction in Glioblastoma With Radiomic Features Using Machine Learning	Baid U, Rane SU, Talbar S, Gupta S, Thakur MH, Moiyadi A and Mahajan A	2020
		Efficacy of Location-Based Features for Survival Prediction of Patients With Glioblastoma Depending on Resection Status	Soltani M, Bonakdar A,Shakourifar N, Babaei R and Raahemifar K	2021
		Multiple Survival Outcome Prediction of Glioblastoma Patients Based on Multiparametric MRI	Wang B, Zhang S, Wu X, Li Y, Yan Y, Liu L, Xiang J, Li D and Yan T	2021
		Radiomics Analysis Based on Magnetic Resonance Imaging for Preoperative Overall Survival Prediction in Isocitrate Dehydrogenase Wild-Type Glioblastoma	Wang S, Xiao F, Sun W, Yang C, Ma C, Huang Y, Xu D, Li L, Chen J, Li H and Xu H	2022
		Survival prediction for patients with glioblastoma multiforme using a Cox proportional hazards denoising autoencoder network	Yan T, Yan Z, Liu L, Zhang X, Chen G, Xu F, Li Y, Zhang L, Peng M, Wang L, Li D and Zhao D	2023
	Hindawi	Tumor Grade and Overall Survival Prediction of Gliomas Using Radiomics	Ye, J., Huang, H., Jiang, W., Xu, X., Xie, C., Lu, B., Wang, X., & Lai, X	2021
	IEEE	Exploring Deep Features from Brain Tumor Magnetic Resonance Images via Transfer Learning	Liu, R., Hall, L. O., Goldgof, D. B., Zhou, M., Gatenby, R. A., & Ahmed, K. B	2016
		Synthetic Minority Image Over-sampling Technique: How to Improve AUC for Glioblastoma Patient Survival Prediction	Liu, R., Hall, L. O., Bowyer, K. W., Goldgof, D. B., Gatenby, R. A., & Ahmed, K. B	2017
		Cox-PASNet: Pathway-based Sparse Deep Neural Network for Survival Analysis	Hao, J., Kim, Y., Mallavarapu, T., Oh, J. H., & Kang, M	2018
		3D Deep Attention Network for Survival Prediction from Magnetic Resonance Images in Glioblastoma	Liu, Z., Sun, Q., Bai, H., Liang, C., Chen, Y., & Li, Z	2019
		CapSurv: Capsule Network for Survival Analysis With Whole Slide Pathological Images	Tang, B., Li, A., Li, B., & Wang, M	2019
		Neuroimaging Based Survival Time Prediction of GBM Patients Using CNNs from Small Data	Ben Ahmed, K., Hall, L. O., Liu, R., Gatenby, R. A., & Goldgof, D. B	2019
		A Novel MKL Method for GBM Prognosis Prediction by Integrating Histopathological Image and Multi-omics Data	Zhang, Y., Li, A., He, J., & Wang, M	2020

Table 2 (continued)

Database	Publisher	Name	Author	Year
		Challenges in predicting glioma survival time in multi-modal deep networks	Aljouie, A., Xue, Y., Xie, M., & Roshan, U	2020
		Cancer Survival Prediction From Whole Slide Images With Self-Supervised Learning and Slide Consistency	Fan, L., Sowmya, A., Meijering, E., & Song, Y	2023
		Prediction of Survival of Glioblastoma Patients Using Local Spatial Relationships and Global Structure Awareness in FLAIR MRI Brain Images	Tran, M., Yang, H., Kim, S., & Lee, G	2023
	IOP science	Interpretable machine learning model to predict survival days of malignant brain tumor patients	Rajput, S., Kapdi, R., Raval, M. S., & Roy, M	2023
	Institute of Physics and Engineering in Medicine	Overall survival time prediction for glioblastoma using multi-modal deep KNN	Tang, Z., Cao, H., Xu, Y., Yang, Q., Wang, J., & Zhang, H	2022
		A subregion-based survival prediction framework for GBM via multisequence MRI space optimization and clustering-based feature bundling and construction	Chen, H., Liu, Y., Pan, X., Yang, Q., Qiang, Y., & Qi, S. X	2023
	Lippincott Williams & Wilkins	Machine Learning Approaches for Prognostication of Newly Diagnosed Glioblastoma	Tunthanathip, T., & Oearsakul, T	2021
	MDPI	Predicting Short-Term Survival after Gross Total or Near Total Resection in Glioblastomas by Machine Learning-Based Radiomic Analysis of Preoperative MRI	Cepeda, S.; Pérez-Nuñez, A.; García-García, S.; García-Pérez, D.; Arrese, I.; Jiménez-Roldán, L.; García-Galindo, M.; González, P.; Velasco-Casares, M.; Zamora, T.; et al	2021
		Prognostic and Predictive Value of Integrated Qualitative and Quantitative Magnetic Resonance Imaging Analysis in Glioblastoma	Verduin, M.; Primakov, S.; Compter, I.; Woodruff, H.C.; van Kuijk, S.M.J.; Ramaekers, B.L.T.; te Dorsthorst, M.; Revenich, E.G.M.; ter Laan, M.; Pegge, S.A.H.; et al	2021
		The Efficacy of Shape Radiomics and Deep Features for Glioblastoma Survival Prediction by Deep Learning	Trinh, D.-L.; Kim, S.-H.; Yang, H.-J.; Lee, G.-S	2022
	Nature	Prognostic models based on imaging fndings in glioblastoma: Human versus Machine	Molina-García, D., Vera-Ramírez, L., Pérez-Beteta, J., Arana, E., & Pérez-García, V. M	2019
		Predicting survival of glioblastoma from automatic whole-brain and tumor segmentation of MR images	Pálsson, S., Cerri, S., Poulsen, H. S., Urup, T., Law, I., & Van Leemput, K	2022
		Risk-aware survival time prediction from whole slide pathological images	Xu, Z., Lim, S., Shin, H., Uhm, K., Lu, Y., Jung, S., & Ko, S	2022
	Oxford University Press	An Online Calculator for the Prediction of Survival in Glioblastoma Patients Using Classical Statistics and Machine Learning	Senders, J. T., Staples, P., Mehrtash, A., Cote, D. J., Taphoorn, M. J. B., Reardon, D. A., Gormley, W. B., Smith, T. W., Broekman, M. L. D., & Arnaout, O	2019
		PathCNN: interpretable convolutional neural networks for survival prediction and pathway analysis applied to glioblastoma	Oh, J. H., Choi, W., Ko, E., Kang, M., Tannenbaum, A., & Deasy, J. O	2021
		Optimizing management of the elderly patient with glioblastoma: Survival prediction online tool based on BC Cancer Registry real-world data	Zhao, R., Zeng, J., DeVries, K., Proulx, R., & Krauze, A	2022

Table 2 (continued)

Database	Publisher	Name	Author	Year
		Comprehensive multimodal deep learning survival prediction enabled by a transformer architecture: A multicenter study in glioblastoma	Gomaa, A., Huang, Y., Hagag, A., Schmitter, C., Höfler, D., Weissmann, T., Breininger, K., Schmidt, M., Stritzelberger, J., Delev, D., Coras, R., Dörfler, A., Schnell, O., Frey, B., Gaipl, U. S., Semrau, S., Bert, C., Hau, P., Fietkau, R., & Putz, F	2024
	RSNA	Radiomic MRI Phenotyping of Glioblastoma: Improving Survival Prediction	Bae, S. J., Choi, Y. S., Ahn, S. S., Chang, J. H., Kang, S., Kim, E. H., Kim, S. H., & Lee, S. K	2018
	SAGE journals	Machine learning based survival prediction in Glioma using large-scale registry data	Zhao, R., Zhuge, Y., Camphausen, K., & Krauze, A	2022
		Pan-Cancer Survival Classification With Clinicopathological and Targeted Gene Expression Features	Zhao, D., Kim, D. Y., Chen, P., Yu, P., Ho, S., Cheng, S. W., Zhao, C., Guo, J. A., & Li, Y. R	2021
		Survival prediction in glioblastoma on post-contrast magnetic resonance imaging using filtration based first-order texture analysis: Comparison of multiple machine learning models	Priya, S., Agarwal, A., Ward, C., Locke, T., Monga, V., & Bathla, G	2021
	SPIE	Integrating Deep Transfer Learning and Radiomics Features in Glioblastoma Multiforme Patient Survival Prediction	Han, W., Qin, L., Bay, C., Chen, X., Yu, K., Li, A., Xu, X., & Young, G. S	2020
	Springer	Prediction of survival with multi-scale radiomic analysis in glioblastoma patients	Chaddad, A., Sabri, S., Niazi, T., & Abdulkarim, B	2018
		Two-Step U-Nets for Brain Tumor Segmentation and Random Forest with Radiomics for Survival Time Prediction	Kim, S., Luna, M., Chikontwe, P., Park, S.H	2020
		Modified MobileNet for Patient Survival Prediction	Akbar, A. S., Fatichah, C., & Suciati, N	2021
		Overall Survival Prediction for Glioblastoma on Pre-treatment MRI Using Robust Radiomics and Priors	Suter, Y., Knecht, U., Wiest, R., Reyes, M	2021
		Combination of pre-treatment dynamic [18F]FET PET radiomics and conventional clinical parameters for the survival stratifcation in patients with IDH-wildtype glioblastoma	Li, Z., Holzgreve, A., Unterrainer, L. M., Ruf, V., Quach, S., Bartos, L. M., Suchorska, B., Niyazi, M., Wenter, V., Herms, J., Bartenstein, P., Tonn, J. C., Unterrainer, M., Albert, N. L., & Kaiser, L	2022
		End-to-End Evidential-Efficient Net for Radiomics Analysis of Brain MRI to Predict Oncogene Expression and Overall Survival	Feng, Y., Wang, J., An, D., Gu, X., Xu, X., & Zhang, M	2022
		Glioblastoma Multiforme Patient Survival Prediction	Rajput, S., Agravat, R., Roy, M., Raval, M.S	2022
	Wolters Kluwer Health, Inc	A Comparative and Summative Study of Radiomics-based Overall Survival Prediction in Glioblastoma Patients	Ruan, Z., Mei, N., Lu, Y., Xiong, J., Li, X., Zheng, W., Liu, L., & Yin, B	2022
MDPI	MDPI	A Predictive Clinical-Radiomics Nomogram for Survival Prediction of Glioblastoma Using MRI	Ammari, S.; Sallé de Chou, R.; Balleyguier, C.; Chouzenoux, E.; Touat, M.; Quillent, A.; Dumont, S.; Bockel, S.; Garcia, G.C.T.E.; Elhaik, M.; et al	2021
		DeepSGP: Deep Learning for Gene Selection and Survival Group Prediction in Glioblastoma	Kirtania, R., Banerjee, S., Laha, S., Shankar, B. U., Chatterjee, R., & Mitra, S	2021
		Ensembles of Convolutional Neural Networks for Survival Time Estimation of High-Grade Glioma Patients from Multimodal MRI	Ben Ahmed, K.; Hall, L.O.; Goldgof, D.B.; Gatenby, R	2022

Table 2 (continued)

Database	Publisher	Name	Author	Year
		Group Lasso Regularized Deep Learning for Cancer Prognosis from Multi-Omics and Clinical Features	Xie, G., Dong, C., Kong, Y., Zhong, J., Li, M., & Wang, K	2019
		Integration of Radiomic and Multi-omic Analyses Predicts Survival of Newly Diagnosed IDH1 Wild-Type Glioblastoma	Chaddad A, Daniel P, Sabri S, Desrosiers C, Abdulkarim B	2019
		Machine Learning and Radiomic Features to Predict Overall Survival Time for Glioblastoma Patients	Chato, L., & Latifi, S	2021
		Multi-Parametric Deep Learning Model for Prediction of Overall Survival after Postoperative Concurrent Chemoradiotherapy in Glioblastoma Patients	Yoon, H. G., Cheon, W., Jeong, S. W., Kim, H. S., Kim, K., Nam, H., Han, Y., & Lim, D. H	2020
		Predicting Overall Survival Time in Glioblastoma Patients Using Gradient Boosting Machines Algorithm and Recursive Feature Elimination Technique	Karami, G.; Giuseppe Orlando, M.; Delli Pizzi, A.; Caulo, M.; Del Gratta, C	2021
		Prognostic Gene Discovery in Glioblastoma Patients using Deep Learning	Wong, K. K., Rostomily, R., & Wong, S. T. C	2019
		Reproducible and Interpretable Machine Learning-Based Radiomic Analysis for Overall Survival Prediction in Glioblastoma Multiforme	Abdulkerim Duman, Xianfang Sun, Solly Thomas, James R. Powell and Emiliano Spezi	2024
Nature	Nature	Clinical measures, radiomics, and genomics offer synergistic value in AI-based prediction of overall survival in patients with glioblastoma	Kazerooni, A. F., Saxena, S., Toorens, E., Tu, D., Bashyam, V., Akbari, H., Mamourian, E., Sako, C., Koumenis, C., Verginadis, I. I., Verma, R., Shinohara, R. T., Desai, A., Lustig, R. A., Brem, S., Mohan, S., Bagley, S., Ganguly, T., O'Rourke, D. M.,... Davatzikos, C	2022
		Multi-Channel 3D Deep Feature Learning for Survival Time Prediction of Brain Tumor Patients Using Multi-Modal Neuroimages	Nie, D., Lu, J., Zhang, H., Adeli, E., Wang, J., Yu, Z., Liu, L. Y., Wang, Q., Wu, J., & Shen, D	2019
		Survival prediction of glioblastoma patients using modern deep learning and machine learning techniques	Babaei Rikan, S., Azar, A. S., Naemi, A., Mohasefi, J. B., Pirnejad, H., & Wiil, U. K	2024
PubMed	American Society of Neuroradiology	Deep Learning of Time–Signal Intensity Curves from Dynamic Susceptibility Contrast Imaging Enables Tissue Labeling and Prediction of Survival in Glioblastoma	Yun, J., Yun, S. Y., Park, J. E., Cheong, E., Park, S. Y., Kim, N., & Kim, H. S	2023
	IEEE	Deep Learning of Imaging Phenotype and Genotype for Predicting Overall Survival Time of Glioblastoma Patients	Tang, Z., Xu, Y., Jin, L., Aibaidula, A., Lu, J., Jiao, Z., Wu, J., Zhang, H., & Shen, D	2020
	Nature	A Deep Learning-Based Radiomics Model for Prediction of Survival in Glioblastoma Multiforme	Lao, J., Chen, Y., Li, Z., Qihua, L., Zhang, J., Liu, J., & Zhai, G	2017
	World Scientific Publishing Company	Deep Integrative Analysis for Survival Prediction	Huang, C., Zhang, A., & Xiao, G	2018
		PAGE-Net: Interpretable and Integrative Deep Learning for Survival Analysis Using Histopathological Images and Genomic Data	Hao, J., Kosaraju, S. C., Tsaku, N. Z., Song, D. H., & Kang, M	2020
Springer	BioMed Central	PASNet: pathway-associated sparse deep neural network for prognosis prediction from high-throughput data	Hao, J., Youngsoon, K., Kim, T., & Kang, M	2018

Table 2 (continued)

Database	Publisher	Name	Author	Year
		Interpretable deep neural network for cancer survival analysis by integrating genomic and clinical data	Hao, J., Youngsoon, K., Mallavarapu, T., Oh, J. H., & Kang, M	2019
		Radiomics for glioblastoma survival analysis in pre-operative MRI: exploring feature robustness, class boundaries, and machine learning techniques	Suter, Y., Knecht, U., Alão, M. N. R. F., Valenzuela, W., Hewer, E., Schucht, P., Wiest, R., & Reyes, M	2020
		Investigating the relevance of major signaling pathways in cancer survival using a biologically meaningful deep learning model	Feng, J., Zhang, H., & Li, F	2021
		neoDL: a novel neoantigen intrinsic feature-based deep learning model identifes IDH wild-type glioblastomas with the longest survival	Sun, T., He, Y., Li, W., Liu, G., Li, L., Wang, L., Xiao, Z., Han, X., Wen, H., Liu, Y., Chen, Y., Wang, H., Li, J., Fan, Y., Zhang, W., & Zhang, J	2021
		Joint learning sample similarity and correlation representation for cancer survival prediction	Hao, Y., Jing, X., & Sun, Q	2022
	Springer	Prognostic factors for survival in adult patients with recurrent glioblastoma: a decision-tree-based model	Audureau, E., Chivet, A., Ursu, R., Corns, R., Metellus, P., Noël, G., Zouaoui, S., Guyotat, J., Reste, P. L., Faillot, T., Litre, F., Desse, N., Petit, A., Emery, E., Lechapt-Zalcman, E., Peltier, J., Duntze, J., Dezamis, E., Voirin, J.,... Pallud, J	2017
		Regression based overall survival prediction of glioblastoma multiforme patients using a single discovery cohort of multi-institutional multi-channel MR images	Sanghani, P., Ang, B. T., King, N. K. K., & Ren, H	2019
		Multi-branch Residual Network Applied to Predict the Three-Year Survival of Patients with Glioblastoma	Fu, X., Chen, C., & Li, D	2020
		Radiogenomics model for overall survival prediction of glioblastoma	Wijethilake, N., Islam, M. & Ren, H	2020
		Survival prediction of patients sufering from glioblastoma based on two-branch DenseNet using multi-channel features	Fu, X., Chen, C., & Li, D	2021
		18F-FET PET radiomics-based survival prediction in glioblastoma patients receiving radio(chemo)therapy	Wiltgen, T., Fleischmann, D. F., Kaiser, L., Holzgreve, A., Corradini, S., Landry, G., Ingrisch, M., Popp, I., Grosu, A. L., Unterrainer, M., Bartenstein, P., Parodi, K., Belka, C., Albert, N., Niyazi, M., & Riboldi, M	2022
		Extracting Radiomic features from pre-operative and segmented MRI scans improved survival prognosis of glioblastoma Multiforme patients through machine learning: a retrospective study	Kaur, G., Rana, P. S., & Arora, V	2022
		Image-based deep learning identifies glioblastoma risk groups with genomic and transcriptomic heterogeneity: a multi-center study	Yan, J., Sun, Q., Tan, X., Liang, C., Bai, H., Duan, W., Mu, T., Guo, Y., Qiu, Y., Wang, W., Yao, Q., Pei, D., Zhao, Y., Liu, D., Duan, J., Chen, S., Sun, C., Wang, W., Liu, Z.,... Zhang, Z	2022

Table 2 (continued)

Database	Publisher	Name	Author	Year
		Prediction of O-6-methylguanine-DNA methyltransferase and overall survival of the patients suffering from glioblastoma using MRI-based hybrid radiomics signatures in machine and deep learning framework	Saxena, S., Agrawal, A., Dash, P. K., Jena, B., Khanna, N. N., Sauseng, P., Kalra, M. M., Višković, K., Fouda, M. M., Saba, L., & Suri, J. S	2023
		Predicting survival in glioblastoma with multimodal neuroimaging and machine learning	Luckett, P. H., Olufawo, M., Lamichhane, B., Park, K. Y., Dierker, D., Verastegui, G. T., Yang, P., Kim, A. H., Chheda, M. G., Snyder, A. Z., Shimony, J. S., & Leuthardt, E. C	2023
		Clinical Feature Ranking Based on Ensemble Machine Learning Reveals Top Survival Factors for Glioblastoma Multiforme	Cerono, G., Melaiu, O., & Chicco, D	2023
Wiley	Wiley	Combining multimodal imaging and treatment features improves machine learning-based prognostic assessment in patients with glioblastoma multiforme	Peeken, J. C., Goldberg, T., Pyka, T., Bernhofer, M., Wiestler, B., Kessel, K. A., Tafti, P. D., Nüsslin, F., Braun, A., Zimmer, C., Rost, B., & Combs, S. E	2019
		Brain tumor segmentation and multiview multiscale-based radiomic model for patient's overall survival prediction	Fiaz, K., Madni, T. M., Anwar, F., Janjua, U. I., Rafi, A., Abid, M. M. A., & Sultana, N	2021
		Brain tumor segmentation and overall survival period prediction in glioblastoma multiforme using radiomic features	Das, S., Das, S., Bose, S., Nayak, G. K., Satapathy, S. C., & Saxena, S	2021
		Multi-level dilated convolutional neural network for brain tumour segmentation and multi-view-based radiomics for overall survival prediction	Rafi, A., Madni, T. M., Janjua, U. I., Ali, M., & Abid, M	2021
		The role of deep learning-based survival model in improving survival prediction of patients with glioblastoma	Moradmand, H., Aghamiri, S. M. R., Ghaderi, R., & Emami, H	2021
		Deep learning and machine learning-based early survival predictions of glioblastoma patients using pre-operative three-dimensional brain magnetic resonance imaging modalities	Kaur, G., Rana, P. S., & Arora, V	2022
		New insights into glioma frequency maps: From genetic and transcriptomic correlate to survival prediction	Bao, H., Ren, P., Yi, L., Lv, Z., Ding, W., Li, C., Li, S., Li, Z., Yang, X., Liang, X., & Liang, P	2022
		Synthetic MRI improves radiomics-based glioblastoma survival prediction	Moya-Sáez, E., Navarro-González, R., Cepeda, S., Perez-Nuñez, A., De Luis-García, R., Aja-Fernández, S., & Alberola-López, C	2022
		Non-invasive prediction of overall survival time for glioblastoma multiforme patients based on multimodal MRI radiomics	Zhu, J., Ye, J., Dong, L., Ma, X., Tang, N., Xu, P., Jin, W., Li, R., Yang, G., & Lai, X	2023

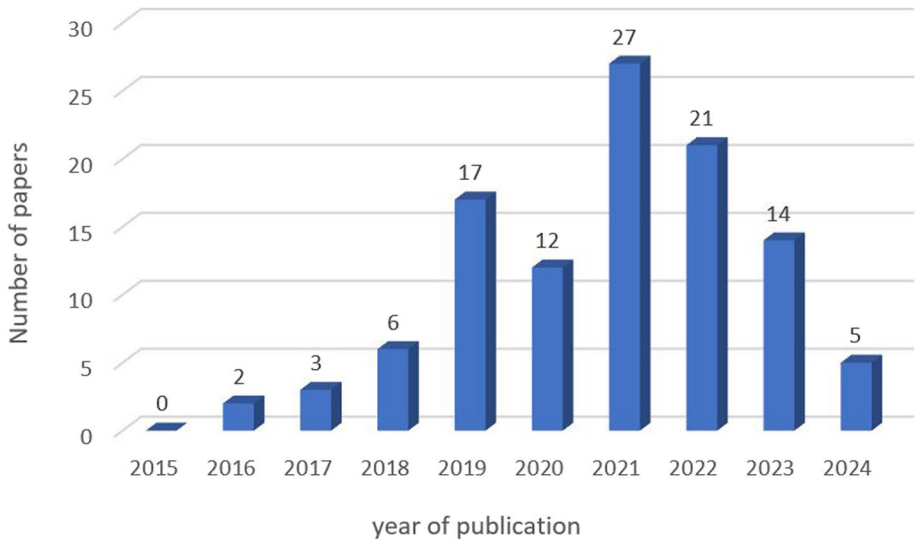


Fig. 3 The number of papers published from 2015 to 2024 in the subject of the study area

Categorization of survival prediction models

The following is a systematic categorization and summary of existing survival prediction models based on statistical, machine learning, and deep learning methods.

- Statistical models:

  Research status: Statistical models, including the Cox proportional hazards model and the Kaplan-Meier estimator, are traditional approaches commonly used for survival analysis. These models are valuable for their interpretability and simplicity, especially in smaller or homogeneous datasets.

  Advantages: These models offer a straightforward interpretation of survival probabilities and hazard ratios, which are clinically useful. They can also work well with limited data and require relatively simple computation.

  Limitations: Statistical models face notable limitations, such as difficulty in managing non-linear relationships and high-dimensional data. They depend on assumptions like proportional hazards, which may not hold true in complex real-world datasets. Furthermore, these models have difficulty in integrating diverse data modalities, such as imaging and molecular features, which are increasingly critical in modern survival prediction [15].
- Machine learning models:
  Research status: ML algorithms like Support Vector Machines (SVM) [16], Random Forests (RF) [17], and Gradient Boosting [18] have gained popularity in survival prediction due to their flexibility and capacity to handle a wide range of data types. These models can accommodate non-linear relationships, making them suitable for clinical data, imaging, and genomic features.
  Advantages: ML models can integrate various data types, such as clinical, radiomic, histological, and molecular features. Their ability to handle complex interactions and capture non-linearities allows them to generate more accurate prognostic models.
  Limitations: Despite their strengths, ML models can be prone to overfitting, particularly with small or unbalanced datasets. Additionally, these models are often less interpretable than statistical models, making it difficult to extract clinically relevant insights. Furthermore, handling missing data and effectively integrating multi-modal datasets, such as clinical, imaging, and molecular data, remain significant challenges, often requiring advanced preprocessing methods [15].
- Deep learning models:
  Research status: DL models, such as Convolutional Neural Networks (CNNs) [19], Recurrent Neural Networks (RNNs) [20], and more recently, Transformer-based models, have shown significant promise in survival analysis. These models are particularly useful when working with high-dimensional and complex data like medical imaging, genomic profiles, and multi-modal clinical data.
  Advantages: DL models are capable of learning intricate patterns and relationships within the data, reducing the need for manual feature engineering. They can seamlessly integrate multiple data modali-

Table 3 Journal disciplines and number of studies in databases

Publisher	Journal/Conference	Number of studies
American Society of Neuroradiology	American Journal of Neuroradiology	2
BioMed Central	BMC Bioinformatics	5
	BMC Medical Genomics	1
	Cancer Imaging	1
British Institute of Radiology	The British Journal of Radiology	1
Cukurova University Faculty of Medicine	Cukurova Medical Journal	1
Elsevier	Advances in Radiation Oncology	1
	Expert Systems With Applications	1
	Clinical Imaging	1
	Computerized Medical Imaging and Graphics	1
	Heliyon	1
	Journal of Biomedical Informatics	1
	Knowledge-Based Systems	1
	Magnetic Resonance Imaging	1
	Neuroscience Informatics	1
	Radiotherapy and Oncology	1
	Surgical Oncology	1
	World Neurosurgery	1
Frontiers	Frontiers in Computational Neuroscience	4
	Frontiers in Oncology	3
	Frontiers in neuroscience	2
Hindawi	Scientific Programming	1
IEEE	2016 International Joint Conference on Neural Networks (IJCNN)	1
	2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)	1
	2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)	1
	2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)	1
	2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)	1
	IEEE Access	2
	IEEE Journal of Biomedical and Health Informatics	1
	IEEE Transactions on Medical Imaging	2
	International Conference on Image Processing (ICIP)	1
IOP science	Machine Learning: Science and Technology	1
Institute of Physics and Engineering in Medicine	Physics in Medicine and Biology	2
Lippincott Williams & Wilkins	International Journal of Nutrition, Pharmacology, Neurological Diseases	1
MDPI	Cancers	7
	Diagnostics	2
	Electronics	2
	Genes	1
	Journal of Personalized Medicine	1
Nature	Scientific Reports	7
Oxford University Press	Bioinformatics	1
	Neuro-oncology Advances	2
	Neurosurgery	1
RSNA	Radiology	1
SAGE journals	Cancer Informatics	1
	Health Informatics Journal	1
	The neuroradiology journal	1
SPIE	Medical Imaging 2020: Image Processing	1

Table 3 (continued)

Publisher	Journal/Conference	Number of studies
Springer	BrainLes 2019: Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries	1
	Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries 2020	1
	European Journal of Nuclear Medicine and Molecular Imaging	1
	European Radiology	1
	International Journal of Computer Assisted Radiology and Surgery	1
	Journal of Medical and Biological Engineering	1
	Journal of Neuro-Oncology	2
	Journal of Healthcare Informatics Research	1
	Lecture Notes in Computer Science	2
	Medical & Biological Engineering & Computing	3
	Multimedia Tools and Applications	1
	Neural Computing and Applications	1
	Proceedings of 2021 International Conference on Medical Imaging and Computer-Aided Diagnosis (MICAD 2021)	1
	Radiation Oncology	1
Wiley	Cancer Medicine	2
	Concurrency and Computation: Practice and Experience	1
	International Journal of Cancer	1
	International Journal of Imaging Systems and Technology	4
	NMR in Biomedicine	1
Wolters Kluwer Health, Inc	Journal of Computer Assisted Tomography	1
World Scientific Publishing Company	Pacific Symposium on Biocomputing 2018	1
	Pacific Symposium on Biocomputing 20,201	1

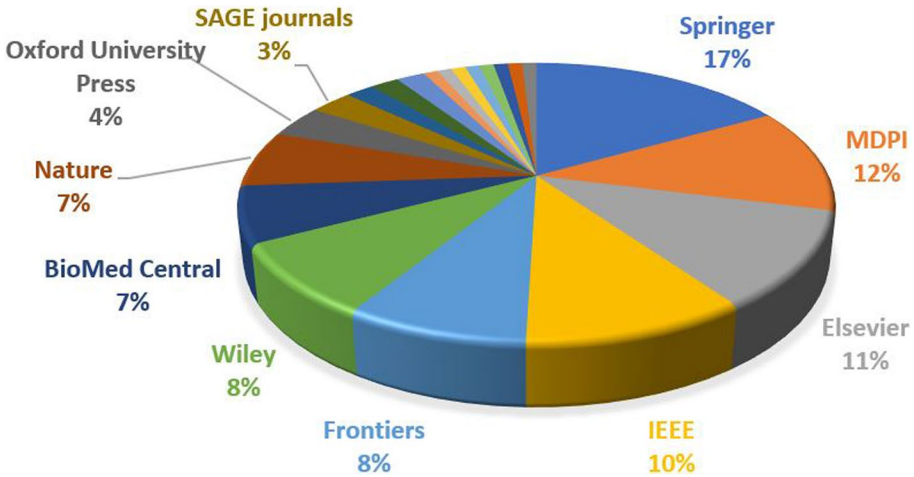


Fig. 4 Pie chart of articles retrieved in different publishers

ties (e.g., imaging, genomic, clinical) and leverage the information for accurate survival predictions.
Limitations: However, DL models require a significant amount of training data, which may not always be available, especially in rare cancer cases. They are computationally intensive and susceptible to overfitting when applied to small datasets. Additionally, their "black-box" nature presents challenges in interpretability, complicating clinical application. To address this, research is focusing on developing explainability methods (e.g., SHAP[1] [21] , LIME[2] [22]) to provide more transparency [15, 23].

Survival prediction as a regression or classification task

Survival time prediction can be approached as either a regression or a classification problem, depending on the task and data type:

- Regression: In regression-based survival prediction, the objective is to estimate a continuous outcome, such as the number of days or months a patient is expected to survive. This approach is suitable when accurate time-to-event data is available.
- Classification: Classification models, on the other hand, predict discrete categories, such as survival status (alive or deceased) or survival groups (e.g., short-term, medium-term, or long-term). This method is often used when the goal is to categorize patients into risk groups for targeted treatments [15, 24].

Table 4 shows the ML and DL methods used in each of the articles and their performance.

Different types of machine learning and deep learning methods are used in different studies. In this research, articles published in journals and conferences of scientific databases have been examined. The results show that random forest and support vector machine are used more than other techniques. Analysis of the publication trends from 2016 to 2024 indicates that traditional models such as Random Forest and Cox Regression were predominant in the early years (2016–2019). This prevalence can be attributed to their interpretability and robust performance, which made them reliable choices for survival prediction. However, from 2020 onwards, there has been a noticeable shift towards the use of neural networks and deep learning models, reflecting the increasing adoption of advanced techniques as computational resources and multi-modal data availability have improved. Despite this shift, Random Forest models continue to be frequently used and remain the most prevalent approach based on the number of publications. Figure 5 shows the types of methods used in different articles.

As can be seen in Table 4, a summary of all articles., researchers used different types of data and various combinations of them to predict survival. Figure 6 shows the most frequent cases. Most of the work has been done using a variety of radiomics and clinical data, followed by radiomics and clinical data.

Most researchers used Python and R to develop survival prediction models. Figure 7 gives the platforms used by the authors to build the machine learning and deep learning models.

The dataset used by the researchers for their research is shown in Fig. 8. It can be seen that among the public datasets, most of the work has been done using TCGA data, followed by BraTS and then TCIA. TCGA, TCIA, and BraTS are three datasets widely used for cancer research and brain tumor segmentation. TCGA stands for The Cancer Genome Atlas, a landmark cancer genomics program that molecularly characterized over 20,000 primary cancers and matched normal samples spanning 33 cancer types. TCGA provides comprehensive genomic, epigenomic, transcriptomic, and proteomic data for researchers to study the molecular basis of cancer and discover new biomarkers and therapeutic targets. https://www.cancer.gov/ccg/research/genome-sequencing/tcga.

TCIA stands for The Cancer Imaging Archive, a service that de-identifies and hosts an extensive archive of medical images of cancer accessible for public download. TCIA organizes the data as collections, typically patients related by a common disease, image modality, or research focus. TCIA also provides supporting data related to the images, such as patient outcomes, treatment details, genomics, pathology, and expert analyses https://portal.gdc.cancer.gov/.

BraTS stands for Brain Tumor Segmentation Challenge, an annual challenge that evaluates state-of-the-art methods for segmenting brain tumors in multimodal magnetic resonance imaging (MRI) scans. BraTS also focuses on predicting overall patient survival. BraTS utilizes multi-institutional pre-operative MRI scans of glioma patients with pathologically confirmed diagnoses and available molecular markers.

Challenges and limitations in survival prediction models

Despite the advancements in survival prediction models, several significant limitations remain that must be addressed to enhance their accuracy and clinical utility. These limitations span multiple dimensions, including data availability, model complexity, data heterogeneity,

[1] Shapley Additive Explanations.

[2] Local Interpretable Model-Agnostic Explanations.

Table 4 Summary of all articles

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Senders et al. [25] **2019**	Clinical features	OS, 1-yr survival status, and compute personalized survival curves	Accelerated failure time (AFT) model based on a log-logistic distribution	R	SEER[a]	C-index = 0.7
Han et al. [26] **2019**	Radiomics features, Deep features	Classification	Elastic Net-Cox proportional hazards	Matlab, R	Brigham and Women's Hospital(BWH), TCGA, TCIA	Log-rank Test Results: BWH: $P < 0.001$ TCGA: $P = 0.14$ (BWH + TCGA): $P = 0.035$
Hao et al. [27] **2023**	mRNA expression, DNA methylation, microRNA expression data	Classification	Deep Neural Network (DNN)	Python	TCGA	Accuracy = 86.2 precision = 90.2 AUC = 85.1
Chaddad et al. [28] **2016**	Geometric shape features	Classification	Random Forests Classifier	Matlab	TCGA, TCIA	mean AUC = 66.99%
Dilek et al. [29] **2021**	Radiomics features	Classification	Linear Discriminant Analysis (LDA)	SPSS, WEKA	Adana Teaching and Research Hospital	AUC: 0.88/0.78, Accuracy: 95%/83.3% (3-/6-month survival)
Fu et al. [30] **2021**	Radiomics features	OS in days	Cox Regression	Matlab	BraTS 2018	C-index = 0.67
Bilmez et al. [31] **2023**	VASARI features, Clinical features	Classification based on 12, 19, and 24 months thresholds	Medium Gaussian SVM	Python, Matlab	Neurosurgery Clinic of Yeditepe University	(Accuracy, Sensitivity, Specificity) 12 months: 96.4%,97.22%,95.55% 19 months: 70.87%,83.02%,60.11% 24 months: 74.66%,67.58%,82.08%
Osman [32] **2019**	Radiomics features, Clinical features	Classification	Ensemble Learning Model	Matlab	BraTS 2018	AUC: 0.81 (short), 0.47 (medium), 0.72 (long); Accuracy: 57.8%
Soltani et al. [33] **2021**	Radiomics features	Survival days Classification	Support Vector Regression (SVR) Artificial Neural Network (ANN)	Python	BraTS 2019	SpearmanR = 0.13, MSE = 26,420.36, median absolute error (MAE) = 133.99, mean absolute error = 139.04, *p*-value = 0.7 Accuracy = 0.78, Precision = 0.6, Sensitivity = 0.8, Specificity = 0.6
Jeong et al. [34] **2019**	MRI- and PET-based volume measures (Radiomics features)	6-month PFS prediction	Cox Regression	Google TensorFlow	Children's Hospital of Michigan	AUC = 0.66, Sensitivity = 0.86, Specificity = 0.63

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Shboul et al. [35] **2019**	Radiomics features, Clinical features	OS in days, Classification	Random Forest Regression (RFR), Extreme Gradient Boosting (XGBoost)	Not Mention	BraTS2017, BraTS2018	RFR BraTS2017: Accuracy = 0.667, MSE = 209,908 Xgboost BraTS2018: Accuracy = 0.679 and MSE = 1,534,663
Wang et al. [36] **2021**	Radiomics features, Clinical features	Predict survival probabilities: OS, PFS and disease-specific survival (DSS)	Cox Proportional hazards	R	TCGA, TCIA	C-index = 0.725, 0.677, and 0.724
Baid et al. [37] **2020**	Radiomics features	OS time in days	Multilayer perceptron (MLP)	Matlab, SPSS	BraTS2018	Accuracy = 0.558, MSE = 338,219.37, MedianSE = 38,408.16, Std. Dev. = 939,986.80, Spearman R = 0.222
Wang et al. [38] **2022**	Radiomics features, Clinical features	OS of IDH-wild-type GBM patients	Multivariate Cox analysis	Matlab, R	Zhongnan Hospital of Wuhan University, TCGA,TCIA	C-index = 0.799
Weninger et al. [39] **2019**	Radiomics features, Clinical features	Survival time in days	Random Foresst Regression (RFR)	Python	BraTS2019	Accuracy = 0.58, p (Binomial) = 0.01, MSE = 144,744, Median err. = 125, SpearmanR = 0.43
Yan et al. [40] **2023**	MRI images,Clinical features	prognostic index (PI)	Convolutional denoising autoencoder (DAE) network combined with the loss function of the Cox proportional hazard regression model	R	BraTS2019, TCGA	C-index = 0.74
Ye et al. [3] **2021**	Radiomics features	Classification	Logistic regression (LR) with dimensionality reduction factor analysis (FA)	Python	BraTS 2017	AUC = 0.63
Liu et al. [41] **2019**	MRI images	Risk value	DeepAttenNet + Regulation (3D deep convolutional neural network using attention mechanism)	Python	Sun Yat-sen University Cancer Center, Guangzhou General Hospital of Guangzhou Military Command, The 3rd Affiliated Hospital of Sun Yat-sen University, TCGA	C-index = 0.68
Zhang et al. [42] **2020**	Gene Expression, CNV, mRNA Expression, histopathological images, clinical features	Classification	Histopathological Integrating Multiple Kernel Learning (HI-MKL)	Python	TCGA	AUC = 0.93221

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Fan et al. [43] **2023**	WSI	Survival time in days	Convolutional Multiple Instance Learning (C-MIL) and an aggregation module	Python	TCGA	C-index = 0.67 MSE = 288 STAGE-5 = 0.595
Tang et al. [44] **2019**	WSI	patient-level survival prediction	CapSurv network (CNN + Capsule Network)	R	TCGA	C-index = 0.67 AUC = 0.722
Aljouie et al. [45] **2020**	SNP, gene expression, MRI images	Classification	multi-path neural network that combines 3D U-Net and ResNet18 and separate multi-layer neural networks	Python	TCGA, TCIA	Accuracy = 0.63
Hao et al. [46] **2018**	Gene expression, Clinical features	Prognostic Index (PI)	Cox-PASNet (Cox + DNN)	Python	TCGA, MSigDB	C-index = 0.6463 ± 0.0332
Liu et al. [47] **2016**	Deep features	Classification	Feature Ranking Classifier	Matlab	H. Lee Moffitt Cancer Center and Research Institute	Accuracy = 0.9545
Ahmed et al. [48] **2019**	Deep features	Classification	small CNN using snapshot ensemble method, VGG16 – Last fully connected layer	Python	TCIA	Acc small CNN = 72.06% Accuracy = 66.18, Specificity = 65.71 Sensitivity = 66.67
Tran et al. [49] **2023**	Global, local features, Radiomics features, Clinical features	OS time in days, Classification	Light GBM (a majority voting technique to decide the survivor class for patient survival classification)	Python	BraTS 2020	Accuracy = 0.621, SpearmanR = 0.576, MSE = 97,216.345
Liu et al. [50] **2017**	Uniform local binary patterns (LBP) histogram features	Classification	Random Forests Classifier	Not mention	Moffitt Cancer Center Dataset, TCGA	AUC = 0.707
Chen et al. [51] **2023**	Geometric features, Radiomics features, Clinical features	Classification, OS time in days	SVC and two targeted sub-regression ensemble models	Python	BraTS 2020	1 year survival prediction: Accuracy = 0.941, Precision = 0.988, Recall = 0.953, F1-score = 0.97, AUC = 0.983; OS prediction: MeanSE = 43 674.49, MeanAE = 88.8, C-index = 0.872
Rajput et al. [52] **2023**	Image-based features, Radiomics features, Clinical features	Survival time in days	RFR	Not mention	BraTS 2020	Accuracy = 0.552 MSE = 79,826.24, medianSE = 14,148.89, stdSE = 148,288, SpearmanR = 0.711

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Tang et al. [53] **2022**	Deep features, Clinical features	Classification	deep KNN	R	BraTS 2019, Institutional	Precision = 0.680 /0.833 /0.600 (S/M/L), Recall = 0.778 /0.654 /0.600 (S/M/L), Accuracy = 0.680
Tunthanathip and Oearsakul [54] **2021**	Clinical features	Classification	Logistic Regression (LR)	R	Tertiary Center of Southern Thailand	Sensitivity = 1, Specificity = 0, PPV = 0.89, NPV = -,Accuracy = 0.84, AUC = 0.82
Cepeda et al. [55] **2021**	Radiomics features, Clinical features	Classification (classify into short-term survivors (< 6 months)/ stratification of patients into low- and high-risk group)	Naive Bayes/ Random Survival Forest	R, Orange	BraTS2020,TCIA, 2 institutional	AUC = 0.769, Accuracy = 0.8, Precision = 0.812, F1 = 0.805/ C-Index = 0.61, IBS = 0.123, iAUC = 0.568, 6m-iAUC = 0.761
Verduin et al. [56] **2021**	VASARI, Radiomics features, Clinical features	OS prediction	Multimodal Cox-Regression	R, Python	Maastricht UMC + and Radboudumc	C-index = 0.711
Trinh et al. [2] **2022**	Deep features, Radiomics features, Clinical features	Survival time in days	MLP	Python	BraTS2018	Accuracy = 57.1%, MSE = 97,531.8, SpearmanR = 0.294
Isson et al. [57] **2022**	Radiomics features, Clinical features	OS,PFS	Random Survival Forest	Python	Copenhagen dataset, BraTS2020	C-index: OS(Copenhagen): 0.670 PFS(Copenhagen): 0.637 OS(BraTS2020): 0.619
Molina-García et al. [58] **2019**	Radiomics features, Clinical features	Prognostic Score (PS)	Optimized Linear Predictive Model (OLPM)	Matlab	Institutional, TCIA	C-index = 0.817
Xu et al. [59] **2022**	WSI	Survival time in days	Two-branch graph convolutional network (GCN)-based model	Python	TCGA	MAE = 303.3, C-index = 0.554
Zhao et al. [60] **2022**	Clinical features	Predicting 6- and 12-month survival probability	Cox proportional hazards regression	R	BC Cancer Registry	C-Index: including all patients: 0.78, only patients who received RT: 0.81
Oh et al. [61] **2021**	mRNA expression, CNV, DNA methylation, Clinical features	Classification	PathCNN	Python	TCGA	AUC = 0.755 ± 0.009
Bae et al. [62] **2018**	Radiomics features, Clinical features, Genetic profiles	OS in days, PFS	Random Survival Forest	R	Institutional	iAUC OS = 0.782, PFS = 0.737

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Zhao et al. [63] **2022**	Clinical features	Risk Score	SVM, Random Forest	Python	BC Cancer Registry	C-Index: 0.738, 0.733 for SVM, RF AUC: 0.821, and 0.825 for SVM, RF
Zhao et al. [64] **2021**	Clinical features, Gene expression	Classification	Random Forest and gradient boosting	Python	TCGA	AUC = 0.76
Priya et al. [65] **2021**	Radiomics features, Clinical features	Classification	Neural Network	R	Not Mention	Accuracy = 0.667, AUC = 0.714, Sensitivity = 0.909, Specificity = 0.167
Ebrahimi Zade et al. [66] **2021**	Clinical features	Survival time in months	Optimal MLP	Python	Not Mention	MAE = 0.087
Wankhede& Selvarani [67] **2022**	Radiomics features	OS in days	Multi-Scale Dense Network (MSDNet)	Python	Kaggle	Accuracy = 0.95
Islam et al. [12] **2021**	Radiomics features, Gene expression	Survival time in days	SVM	Python	TCGA	MSE = 11,731, Accuracy = 84.62%
Han et al. [68] **2020**	Radiomics features, Deep features	Survival score	Elastic net-Cox model proportional hazard	Not Mention	TCGA, TCIA, Brigham and Women's Hospital(BWH)	The model achieved statistically significant results (p-value < 0.05)
Li et al. [69] **2022**	Radiomics features, Clinical features	Classification	Logistic regression (LR)	Python	Department of Nuclear Medicine of the LMU Munich	AUC = 0.74, Accuracy = 64.3%, Sensitivity = 50%, Specificity = 70% PPV = 40%, NPV = 77.8%
Feng et al. [70] **2022**	Radiomics features, Clinical features	Classification	End-to-End Evidential-Efficient Net that incorporates EDL layers as a classifier	Python	BraTS2019	Accuracy = 0.513, MSE = 79,265
Rajput et al. [71] **2021**	Image-based features, Radiomics features	Classification	Gradient Boosting Regressor (GBR)	Not mention	BraTS2020	Accuracy = 0.621, MSE = 141,065.30, medianSE = 23,528.40, stdSE = 236,728.70, SpearmanR = 0.338
Akbar et al. [72] **2021**	Deep features, clinical features	Survival time in days	MobileNet combined with a linear survival prediction model (SPM)	Python	BraTS2020	Accuracy = 0.402, MSE = 470,903.86 MedianSE = 1, 79,348.037, StdSE = 1,393,608.343, SpearmanR = 0.086

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Suter et al. [73] **2021**	Radiomics features, Clinical features	Classification	Random Forest Regression	Python	BraTS2020	Balanced acc. = 0.512, Accuracy = 0.515, MSE = 112,653, Spearman's rho = 0.438
Chaddad et al. [74] **2018**	Radiomics features, Clinical features	Predict PFS and OS	Random Forest Regression	Matlab	TCGA, TCIA	AUC: PFS: 85.37 OS: 85.54
Kim et al. [75] **2020**	Radiomics features	Survival time in days	Random Forest Regressor	Python	BraTS2019	Accuracy = 50.5%, MSE = 409,680.9, medianSE = 50,625
Ruan et al. [76] **2022**	VASARI, Radiomics features, Clinical features	Survival probability	Random Survivsl Forest	R	TCGA, Institutional	C-index = 0.935, IBS = 0.169
Fathi Kazerooni et al. [77] **2022**	MGMT methylation, Genomics, Radiomics features, Clinical features	Risk Stratification	Cox proportional hazards (Cox-PH)	R	University of Pennsylvania (UPenn)	C-index = 0.75, IBS = 0.086, IBS diference* (%) = −24.8
Nie et al. [78] **2019**	Deep features, Radiomics features, Clinical features	Classification	CNN + SVM	Not mention	Huashan hospital, Shanghai, China	Accuracy = 90.66, Sensitivity = 96.77, Specificity = 85.04, PPR (%) = 85.82, NPR (%) = 96.31
Yun et al. [79] **2023**	Perfusion patterns, Clinical features	OS of patients with IDH wild-type glioblastoma	Cox proportional hazard regression	R	Asan Medical Center	C-index = 0.72
Tang et al. [80] **2020**	Deep features, Genomic biomarkers,Clinical features	OS time in days	A multi-task convolutional neural network (CNN)	Python	single-center GBM biobank database collected from a collaborated hospital in China	RMSE = 177, Pearson correlation coefficient = 0.4695
Lao et al. [81] **2017**	Radiomics features, Clinical features	Risk score	LASSO Cox regression model	R	Sun Yat-Sen University Cancer Center,TCGA,TCIA	C-Index = 0.739
Huang et al. [82] **2018**	WSI, CNV	Risk score	Deep survival learning model (CNN-based and one FCN-based subnetwork)	Not mention	TCGA	C-Index = 0.6045
Hao et al. [83] **2020**	WSI, gene expression, Clinical features	Prognostic Index (PI)	PAGE-Net (CNN + Cox)	Python	TCGA, TCIA	C-Index = 0.702
Ghanem et al. [84] **2023**	Clinical features	months of survival and vital status	Probabilistic matrix factorization	Python	SEER (Surveillance Epidemiology and End Results)	C-index = 0.6918, the Brier score (BS) = 0.0934, integrated BS (IBS) = 90.66%
Su et al. [85] **2021**	Deep features	Classification	SVM	Python	TCGA	Accuracy = 94.8, Sensitivity = 93.3, Specificity = 96.7, AUC = 0.98

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Lu et al. [86] **2020**	VASARI, Radiomics features, Clinical features	Risk score	Random Survival Forest (RSF)	R	Institutional	iAUC = 96.2 ± 1.7 and C-index = 90.0 ± 0.3
Sanghani et al. [87] **2018**	Radiomics features, Clinical features	Classification	SVM	Python	BraTS2017	Accuracy: 2-class = 98.7%, 3-class = 88.95%
Kim et al. [88] **2023**	Clinical features	Risk score (OS and PFS(	Random Survival Forest (RSF)	R	Yonsei Cancer Center	C-index = 0.72 and IBS = 0.12 for OS prediction; C-index = 0.70 and IBS = 0.12 for PFS prediction
Suter et al. [89] **2020**	Radiomics features, Clinical features	Classification	MLP	Python	BraTS 2018, Institutional	single center: AUC = 1.00, Bal. Accuracy = 95%, Accuracy = 95%, Specificity = 93%, Sensitivity = 97%, F1 = 0.95, Precision = 95% multi center: AUC = 0.51, Bal. Accuracy = 55%, Acc uracy = 43%, Specificity = 30%, Sensitivity = 80%, F1 = 0.43, Precision = 29%
Hao et al. [90] **2019**	Gene expression, Clinical features	Prognostic Index (PI)	Cox-PASNet (Cox + DNN)	Python	TCGA	C-index = 0.6347
Feng et al. [91] **2021**	Gene expression, CNV, Clinical features	Survival time	DeepSigSurvNet (CNN based)	Python	TCGA	C-index = 0.6274
Hao et al. [92] **2022**	DNA methylation, mRNA expression, miRNA expression data	Survival risk	Multi-type Data Joint Learning (MDJL)	Python	TCGA	C-index = 0.875
Sun et al. [93] **2021**	Clinical features, neoantigen intrinsic features	Classification IDH wild-type GBMs	NeoDL (LSTM based)	R, Python	TCGA	AUC = 0.988, p value < 0.0001
Hao et al. [94] **2018**	Gene expression, pathways	Classification	Pathway-Associated Sparse Deep Neural Network (PASNet)	Python	TCGA, MSigDB	AUC = 0.6622, F1 score = 0.3978
Wiltgen et al. [95] **2022**	Radiomics features, Clinical features	Classifcation, Survival time	RF	Python	Department of Nuclear Medicine, University Hospital of Munich	C-index = 0.76
Kaur et al. [96] **2022**	Radiomics features, Clinical features	Survival time in days	Ridge Regression	Python	BraTS2020	Accuracy = 0.690, MSE = 83,666.22, medianSE = 16,620.98, stdSE = 161,480.20, SpearmanR = 0.635

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Yan et al. [97] **2023**	MRI images	OS risk score	CNN based on the ResNet-34 architecture	Python	five institutions: FAHZZU,NFHNFMU, HPH, TAHSYSU, and GGHGMC and three public databases: TCIA, TCGA, and Chinese Glioma Genome Atlas (CGGA)	C-index of 0.80 for the internal validation set, 0.73 for the external test set 1, 0.76 for the external test set 2, and 0.73 for the public test set
Fu et al. [98] **2020**	Radiomics features	Classification	a multi-branch residual network survival prediction (M-BSP)	Python	TCIA	Accuracy = 0.89, Recall = 0.83, Precision = 0.94, F1 score = 0.88, AUC = 0.878
Saxena et al. [99] **2023**	Radiomics features	Classification	SVM	Not Mention	BraTS2019	Accuracy = 0.70, Spec = 0.66, AUC = 0.78, Sens = 0.74, Precision = 0.69
Audureau et al. [100] **2017**	Clinical features	OS from progression	Conditional Random Forest	Stata, R	multicentric retrospective study on the behalf of the Club de Neuro-Oncologie of the Société Française de Neurochirurgie	C-index = 70.37%
Wijethilake et al. [101] **2020**	Radiomics features, gene expression, Clinical features	OS time in days	Linear regression (LR)	Python	TCGA, TCIA	MSE = 7576.36, Accuracy = 91.67%
Sanghani et al. [102] **2019**	Radiomics features, Clinical features	OS time in days, Classification	Support Vector Regression (SVR)	Python	BraTS2017	MPE = 14.6, 95% CI = 195.8, $R^2 = 0.92$ The 2-class prediction Accuracy = 86%, 3-class prediction Accuracy = 80%
Fu et al. [103] **2021**	MRI images	Classification	Two-branch DenseNet survival prediction model	Not Mention	TCIA	Accuracy = 0.940 Recall = 0.920 Precision = 0.958 F1 score = 0.938 AUC = 0.942
Fiaz et al. [104] **2021**	Radiomics features	OS time in days	Random Forest Regression	Python	BraTS 2020	Accuracy = 0.483, MSE = 62,620.41 Median SE = 34,556.951 Spearman R = 0.552

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Das et al. [105] **2021**	Radiomics features	Classification	SVM	Python	BraTS 2017	Accuracy: 2-Class: 0.70 3-Class: 0.66
Kaur et al. [106] **2022**	Deep features, Imaging-based features, Clinical features	Survival time in days	Automated regression-based ML	Python	BraTS 2020	MSE = 87 067.33, medianSE = 30,915.66, stdSE = 124 721.48,SpearmanR = 0.326
Rafi et al. [107] **2021**	Radiomics features, Clinical features	OS time in days	Random Forest Regression	Python	BraTS 2019	Accuracy = 48.3%, MSE = 92,599.598, MedianSE = 31,538.806, stdSE = 142,490.975,SpearmanR = 0.252
Bao et al. [108] **2023**	Radiomics features, Clinical features	Classification	nonlinear (radial) kernel-based SVM	R	HMU Cancer Hospital, TCIA, TCGA	Accuracy = 0.71
Zhu et al. [109] **2023**	Radiomics features, Clinical features	OS time in days	Support Vector Regression (SVR)	Python	BraTS2020, BraTS2019, BraTS2018	BraTS2020: MSE = 139,571.9641, MAE = 254.6866, RMSE = 360.8906. BraTS2019: MSE = 121,095.8173, MAE = 237.6304, RMSE = 337.3586. BraTS2018: MSE = 132,036.0251, MAE = 241.9858, RMSE = 350.1011
Moya-Sáez et al. [110] **2022**	Radiomics features	Classification	Extreme gradient boosting (XGB)	Matlab	BraTS2020, TCIA, Hospital Universitario Río Hortega, Valladolid, Spain, and Hospital Universitario 12 de Octubre, Madrid, Spain	It is not reported numerically in the text of the article
Moradmand et al. [111] **2021**	Genomics features, Clinical features	OS prediction	DeepSurv (Cox-based deep feed forward neural network)	R,Python	TCGA, IVY GAP	C-index = 0.7
Ammari et al. [112] **2021**	Radiomics features, Clinical features	Classification	Random Forest Classifier, Logistic Regression	Python	BraTS2019	AUC = 0.85, 0.74, 0.58,0.71 for survival at 9, 12,15 and 22 months, respectively
Kirtania et al. [113] **2021**	RNA-seq data	Classification	MLP	Python	TCGA	Accuracy = 0.89, AUC = 0.97
Ben Ahmed et al. [114] **2022**	MRI images	Classification	Multi-branch 3D convolutional neural network	Python	BraTS 2019	Accuracy = 74%

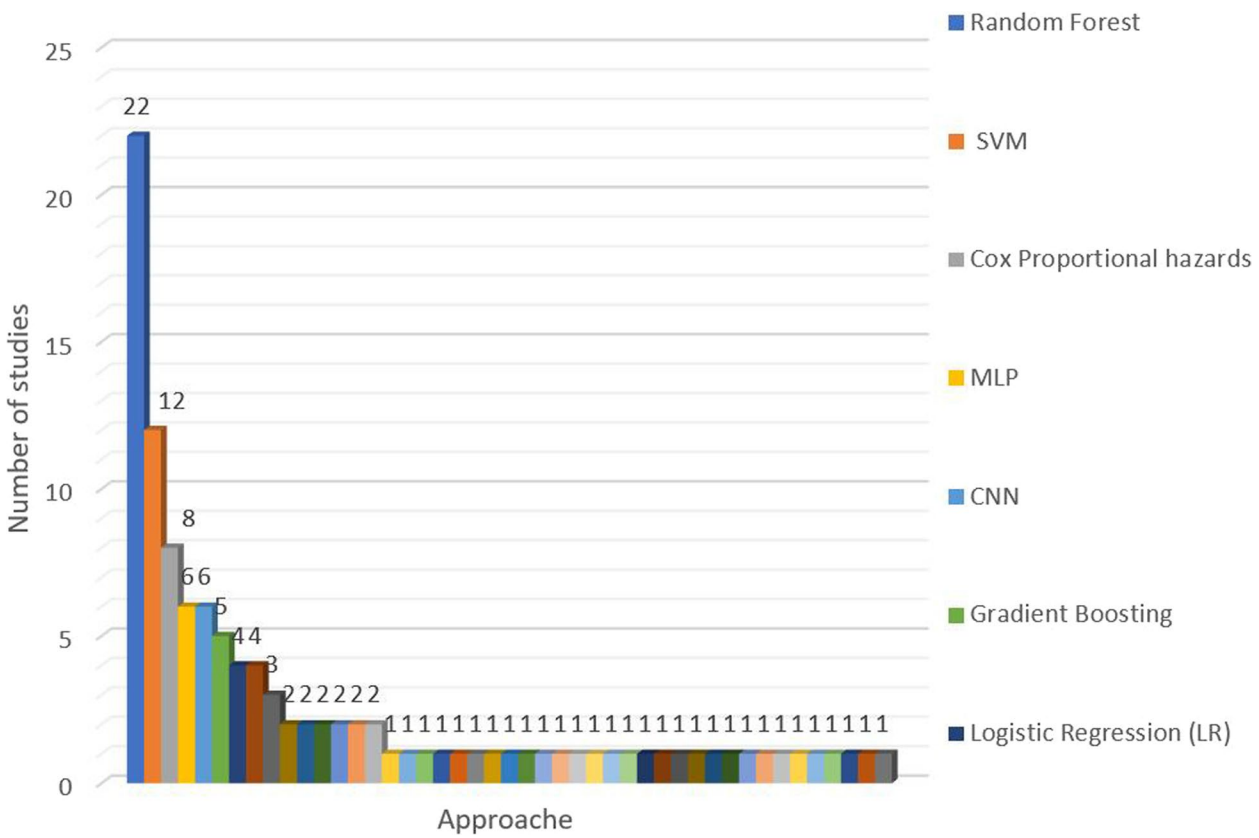
Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Xie et al. [115] **2019**	Gene expression, CNV, Protein expression, DNA somatic mutation and clinical features	Survival time of patients	GDP (Fully connected deep learning framework and Cox proportional hazard)	R, Python	TCGA	C-index > 0.8
Chaddad et al. [116] **2019**	Radiomics features, Clinical features,Genetic, Protein expression	Classification	Random Forest Classifier	Matlab	TCGA, TCIA, McGill University Health Centre	AUC = 78.24%
Chato & Latifi [117] **2021**	Radiomics features, Clinical features	Classification	Neural Network	Not mention	BraTS 2019	Accuracy = 65.1%
Yoon et al. [118] **2020**	Radiomics features, Clinical features	OS time prediction	CNN–based deep learning model	Google Tensorflow, R	Samsung Medical Center (Seoul, Korea)	C-index = 0.768, iAUC = 0.790
Karami et al. [119] **2021**	Image-based features, Clinical features	Classification	Random Forest-Recursive Feature Elimination (RF-RFE) GBoost	Python, Matlab	ITAB, Università degli Studi "G. d'Annunzio" Chieti-Pescara	Accuracy = 0.75, Precision = 0.75, Recall = 0.74, F1-Measure = 0.75, AUC = 0.741, MCC = 0.48
Wong et al. [120] **2019**	Gene expression, Clinical features	Survival risk	Deep MLP	Python	TCGA	C-index = 0.69
Duman et al. [121] **2024**	Radiomics and Clinical features	Classification	regularized Cox regression(Cox-Lasso)	Python	Brats2020, STORM_GLIO	C-Index: 0.69 iAUC: 0.81
Gomaa et al. [122] **2024**	MR images, clinical, and molecular-pathologic data	an individual patient's survival function throughout the follow-up period	Transformer-based deep learning model	Not Mention	UPenn-GBM, UCSF-PDGM, (RHUH)-GBM	C-index = 0.707 (UPenn), 0.672 (UCSF), 0.618 (RHUH)
Babaei Rikan et al. [123] **2024**	Clinical features	Classification and Regression(the number of months)	Feed-Forward Deep Neural Network (DNN)	Python	SEER	Classification: Holdout: Accuracy = 90.25%, Cross-validation: Accuracy = 90.22%. Regression: Holdout: $R^2 = 0.6565$, Cross-validation: $R^2 = 0.6622$
Sun, J., & Peng, Y. [124] **2024**	Clinical, Radiomics, and Imaging (MRI) features	Classification	Dual-Graph Neural Network (GNN)	Python	BraTS2020	Accuracy = 0.586
Lyu et al. [125] **2024**	Imaging Features	Classification	SurvNet	Python	TCGA-GBM	Accuracy = 0.824 Precision = 0.889 Recall = 0.778 F1-score = 0.822 AUC = 0.931
Luckett et al. [126] **2023**	Clinical data, Contralesional cortical thickness (CT) measures, Functional connectivity (FC) features	Classification	DNN	MATLAB, R	Washington University Medical Center	Accuracy = 0.906

Table 4 (continued)

Author	Type of Data	Task	Approach	Programming Languages	Dataset	Results
Cerono et al. [127] **2023**	Clinical features, Genomic features, Tumor characteristics	Survival time of patients	DeepSurv	Python	Shieh, Berendsen, Lammer datasets	Shieh dataset:C-index = 0.583, Berendsen dataset: C-index = 0.776, Lammer dataset C-index = 0.64

[a] Surveillance, Epidemiology and End Results


Random Forest
SVM
Cox Proportional hazards
MLP
CNN
Gradient Boosting
Logistic Regression (LR)
Number of studies
Approache
0
5
10
15
20
25
22
12
8
6 6
5
4 4
3
2 2 2 2 2 2
1 1 1

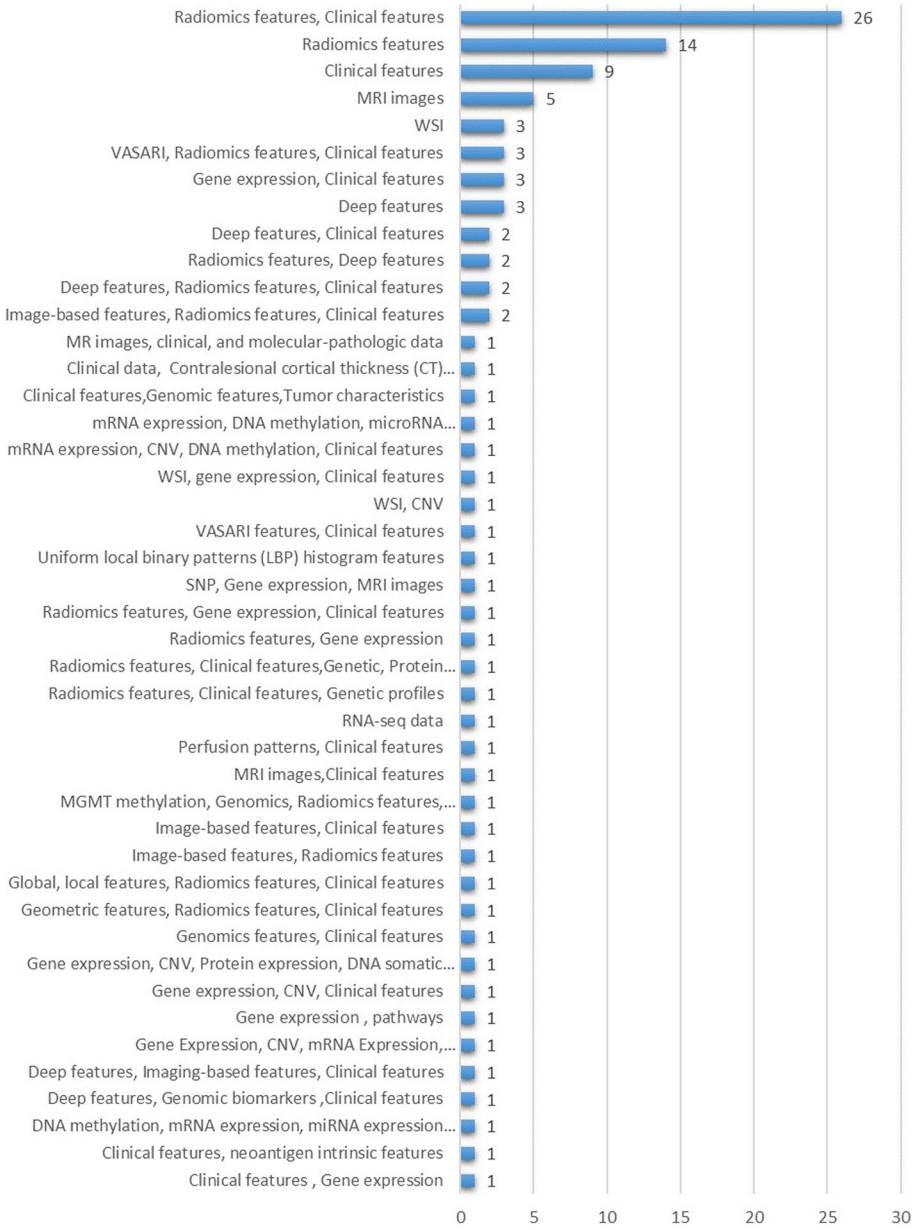


Fig. 6 Number of retrieved articles by data type

Data heterogeneity

The variability in clinical datasets presents a major challenge for survival prediction models. This heterogeneity arises from several factors:

Variability Among Patients: Datasets often include data from patients with diverse cancer types, stages, and treatment histories. While this diversity can aid in creating more generalizable models, it also introduces significant variability. For instance, a dataset might contain more glioblastoma cases than rarer types like oligodendroglioma, leading to imbalanced data that models may struggle to generalize across. Techniques such as oversampling, undersampling, and synthetic data generation (e.g., SMOTE[3]) [130] can be used to address these imbalances.

Integration of Multimodal Data: Combining diverse data types (e.g., imaging, genomics, and clinical records) offers a holistic view of patient health but introduces complexity in data integration. Not all patients may have complete data across all modalities, complicating the

[3] Synthetic Minority Over-sampling Technique.

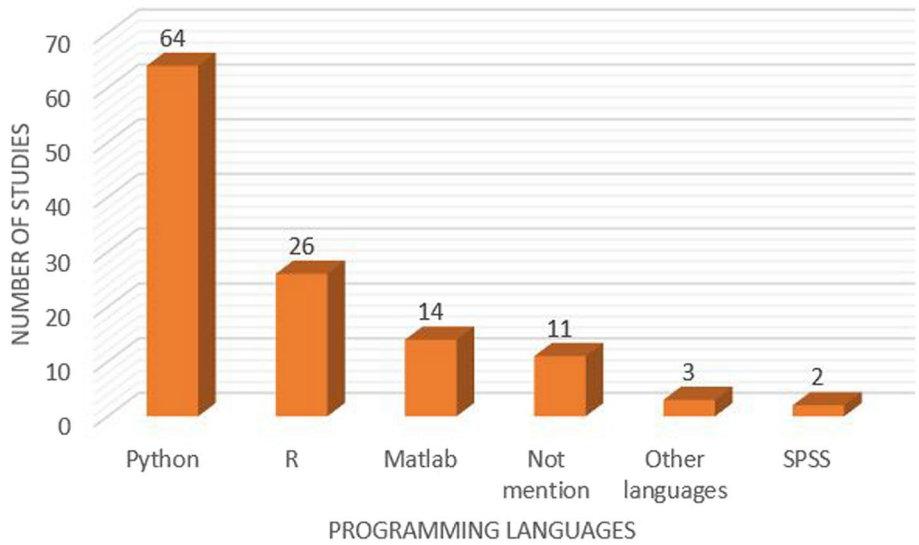


Fig. 7 Programming languages used in articles

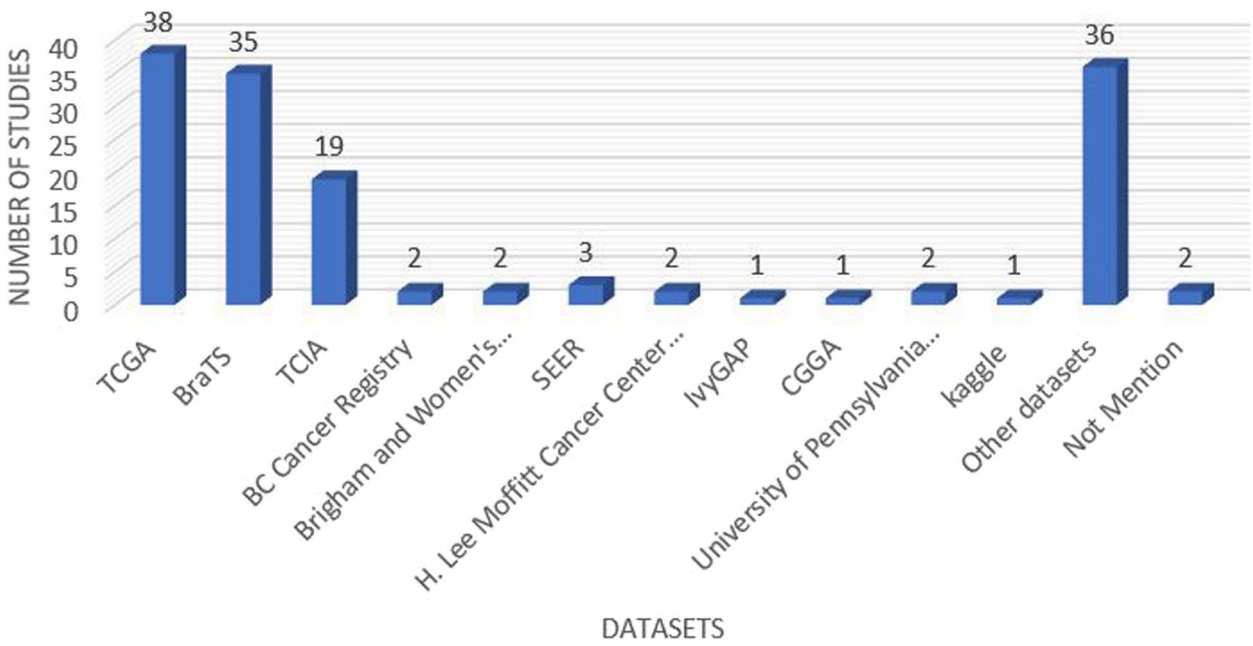


Fig. 8 Categorizing of articles based on various datasets

development of models that can effectively leverage the available information. Advanced methods like multi-view learning or deep learning architectures that can handle missing modalities may be required to address this challenge.

Handling Missing Data: Missing data is common in clinical datasets due to incomplete tests or patient dropouts. If not managed properly, this missing information can bias model outcomes. Techniques such as K-Nearest Neighbors (KNN) imputation, multiple imputation strategies, or Bayesian models can help manage this issue, but their application needs careful consideration to avoid introducing new biases.

Variations in Data Collection Methods: Inconsistencies in data collection methods, such as differences in imaging devices or sequencing platforms, can affect data quality. For example, MRI scans from machines with different magnetic field strengths (e.g., 1.5T vs. 3T) may produce

images of varying quality, impacting model performance if these discrepancies are not harmonized. Standardization techniques, such as ComBat for genomic data or intensity normalization for imaging, are essential for ensuring consistency and reliability [11].

Computational complexity and resource requirements

Advanced deep learning models, while effective in capturing complex patterns, require significant computational resources for training and deployment. These resource-intensive requirements may not be feasible in all research settings, particularly in low-resource environments. Additionally, the complexity of these models adds to the challenge of making them accessible and practical for widespread clinical use [131].

Bias and ethical considerations

Bias in datasets and models is another critical issue that can affect the fairness and generalizability of survival prediction models. Existing models may reflect and reinforce biases present in healthcare data, such as demographic disparities or unequal access to healthcare services. Ensuring that models are developed and validated on diverse and representative datasets is crucial to prevent such biases. Moreover, ethical concerns related to the use of sensitive data, particularly genetic and genomic information, must be addressed. Data protection measures, including secure storage and anonymization techniques, are essential for maintaining patient privacy and ensuring the responsible use of AI in healthcare [132].

Discussion

Survival prediction is a key application of computational techniques in healthcare, which aims to improve the quality of care for patients. It is a type of time-to-event prediction, which measures the duration from the start of observation until the occurrence of death (or another specific event) for the patient under observation, or until the end of observation. Survival analysis can help to estimate the risk of death or recurrence for patients based on various types of data, such as clinical, imaging, omics, and environmental data. In the context of cancer survival, survival analysis corresponds to the time between diagnosis and death from the disease, and can help to assess the prognosis and personalized treatment planning for cancer patients. Survival prediction is a difficult task that requires the combination of multiple types of information to capture the diversity and variability of the disease. Traditional statistical methods are not sufficient or suitable to predict survival from imaging, multi-omics, or their integration, as they have limitations in dealing with high-dimensional and non-linear. Therefore, computational approaches that use artificial intelligence techniques, such as machine learning (ML) and deep learning (DL), have been developed and applied to enhance the precision and robustness of survival prediction models.

The results of this systematic review show that machine learning and deep learning methods have been widely applied to survival prediction of glioblastoma patients, using different types of input data, such as clinical features, molecular markers, imaging features, radiomics features, omics data, or a combination of them. This systematic review examines the data sources, methods, performance measures, and results of 107 studies published from 2015 to 2024. Throughout this review, we noticed that:

1. The most frequently used method was random forest, which is a tree-based ensemble learning method that can handle high-dimensional and heterogeneous data.
2. The most prevalent input data was a combination of radiomics and clinical data, which can capture both the tumor phenotype and the patient characteristics, and provide complementary information for survival prediction.
3. Most researchers used Python and R to develop survival prediction models.
4. Among the public datasets, most of the studies has been done using TCGA data.

Future work

- Exploration of transformer models and new-generation AI techniques:
    - Current Research Landscape: While there are a few studies that have started exploring the application of transformer models for survival prediction, this area is still relatively new and underdeveloped in the context of medical-assisted diagnosis. Recent advancements in AI demonstrate the potential of transformers, particularly in handling complex, multi-modal data integration. Transformer architectures, such as Vision Transformers (ViTs) and other variants, have shown promise in other fields, and their application is beginning to emerge in survival analysis.
    - Proposed Future Direction: Future studies should build on these initial findings and further develop transformer-based models for survival prediction. Given their ability to process sequential and high-dimensional data effectively, transformers have the potential to significantly enhance the accuracy of

models that integrate imaging, clinical, and genetic data.

- Integration of big data, multi-omics, and multi-modality:
  - The Role of Big Data: The next generation of medical-assisted diagnosis models must leverage big data to capture the complexity and diversity of patient populations. This involves incorporating large-scale datasets that encompass patient demographics, clinical records, imaging data, and genetic profiles from multiple institutions and populations.
  - Multi-Omics Data: Integrating diverse omics data (e.g., genomics, transcriptomics, proteomics) can provide deeper insights into the molecular mechanisms driving cancer progression and patient outcomes. Future models should develop strategies to efficiently combine these different data types, using advanced techniques like multi-omics integration frameworks or hierarchical models that account for data at different biological levels.
  - Modality Data: The future of medical diagnosis models lies in effectively merging various data modalities (e.g., MRI, CT scans, histopathological images, clinical data, and omics information) to create a unified predictive model. Developing deep learning architectures, such as hybrid networks that integrate CNNs, RNNs, and transformers, could facilitate the processing and integration of these data types.
  - Large Models: Implementing large-scale models capable of processing and learning from extensive and diverse data is crucial for improving prediction accuracy. Although transfer learning is currently underexplored in survival prediction, it presents an opportunity for future research. Applying transfer learning techniques could help build models that adapt to new data efficiently, especially in scenarios where acquiring large amounts of labeled medical data is challenging.
  - Development of Ethical and Transparent AI Systems:

- Development of ethical and transparent AI systems:
  - Addressing the "Black Box" Problem: As AI models become more complex, enhancing their interpretability remains critical. Future work should prioritize the development of Explainable AI (XAI) solutions that provide clinicians with transparent and understandable outputs. These models must offer insights into which features or data sources influence predictions, enabling clinicians to make informed decisions confidently.
  - Privacy and Security: With the integration of big data and multi-omics information, ensuring patient privacy and data security is essential. Research should focus on developing privacy-preserving techniques such as federated learning and encrypted data sharing, allowing models to learn from distributed datasets without compromising patient confidentiality.

- Scalability and generalization:
  - Future research should explore ways to scale these advanced models for global application, ensuring that they generalize across diverse populations and healthcare systems. Collaborations across multiple research centers and hospitals will be necessary to gather and validate multi-center datasets that enhance the models' robustness and applicability.

Conclusion

This article has presented a systematic review of the current state-of-the-art machine learning and deep learning methods for survival prediction of glioblastoma patients. The article has analyzed the data sources, methods, performance metrics, and outcomes of 107 studies published from 2015 to 2024. The article revealed that random forest was the most frequently used method, and combining radio mics and clinical data was the most prevalent input data. Predicting survival with high accuracy can assist personalized clinical decision-making in glioblastoma patients. The scientific domain is shifting towards machine learning and deep learning models built on high-dimensional data, often with promising results. However, these models have yet to be implemented in clinical care. To facilitate the clinical adoption of high-performing machine learning and deep learning models, future efforts should focus on harmonizing data acquisition methods, improving model interpretability, and externally validating these models in a multicentered, prospective way. The article hopes to provide a helpful reference and guidance for researchers and practitioners interested in this topic.

Abbreviations

ANN	Artificial Neural Network
CNN	Convolutional Neural Network
CSS	Cancer-specific survival
DFS	Diseasefree survival
DL	Deep Learning
DNN	Deep Neural Network
GBM	Glioblastoma Multiforme

HGG	High Grade Gliomas
LGG	Low Grade Gliomas
ML	Machine Learning
MRI	Magnetic Resonance Imaging
OS	Overall Survival
PET	Positron emission tomography
PFS	Progression-free survival
PRISMA	Preferred Reporting Items for Systematic reviews and Meta-Analysis
SVM	Support Vector Machine
RF	Random Forest
TMZ	Temozolomide
WHO	World Health Organization

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12885-024-13320-4.

Supplementary Material 1.

Acknowledgements
Not applicable.

Authors' contributions
R.P. performed the literature search and data analysis, and R.P. and A.A.S and M.M. drafted and critically revised the work.

Funding
Not applicable.

Data availability
The data supporting the findings of this study are available from previously published articles and are accessible from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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Received: 9 March 2024 Accepted: 10 December 2024
Published online: 27 December 2024

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